Areg Hovumyan

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Education

California State Polytechnic University, Pomona — Pomona, CA
Bachelor of Science in Computer Engineering — GPA: 3.6/4.0 — *Expected Dec. 2027*

- **Relevant Coursework:** Digital Logic Design, Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms

Relevant Experience

NASA L'SPACE Mission Concept — May 2026 – Present
Electrical Member - Robotic Lunar Lander Concept — *Remote*

- **Presented the CDH progress to NASA L'SPACE** members to evaluate and consider for an actual mission.
- **Designed the power subsystem architecture**: solar battery selection and power budget allocation across the subsystems.
- Developed **MCUs, regulators, and AC-DC converters** by calculating for radiation tolerance and power draw.
- Built**Command & Data Handling (C&DH) architecture**, including communication between the earth and the moon.
- Worked alongside the science, thermal, and mechanical subteams as an electrical subsystem member.

Project Experience

Software Engineering Team Lead: Autonomous Drone — Jun 2025 – Present
*Sponsored by **Lockheed Martin*** — *Pomona, CA*

- Collaborated with the Electrical and Payload teams to ensure safe and successful **flight deployments.**
- Presented **Lockheed Martin engineers** the **system design** during a **Preliminary Design Review (PDR)**.
- Developed an **object recognition pipeline** on a **Jetson**, achieving **3x quicker detection** and increased accuracy to **99%**.
- Implemented**multi threading in C++/Python** using **lock-protected shared state** .
- **Improved runtime performance and cross-thread data consistency by 1.5x.**

Embedded Systems Team Lead: Autonomous Ground Vehicle — Jan. 2026 – Present
Northrop Grumman Collaboration Project — *Pomona, CA*

- Presented **telemetry design** to **Northrop Grumman engineers** during a **Preliminary Design Review (PDR)**.
- Created **reusable device drivers** and **register-level C code** for micro controller peripherals.
- Used **logic analyzers** to debug problems that cross **electrical, firmware, and software borders** .
- Creating and evaluating **multi-agent autonomy behaviors** in **actual flight conditions**.

Software Engineer: Autonomous Vehicle Lab — Apr 2026 – Present
University Research Lab — *Pomona, CA*

- Implemented **Teensy, STM32**, and other microcontrollers for controlling embedded vehicle systems.
- Designed a **manually controllable and autonomous rover** for a competition.
- Developed **CI/CD pipelines** for the team to push safer code to production.
- Created **unit tests using GoogleTest and pytest** to improve system reliability.
- Utilized AI agents to refactor, rewrite, improve, and test the software.

Full-Stack Software Engineer/Founder — May 2025 – Dec 2025
Web development company founder — *Remote*

- Lead a team of 10 engineers to build full-stack websites for various clients.
- Created Websites that include chatbots, Interactive interfaces, futuristic designs, and proper databases.
- Created safety guardrails, debugging pipelines, and monitoring tools to improve the reliability of all the products.
- Hired engineers from different disciplines such as systems, web, , HR, and DevOps.

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript, Verilog, VHDL, ARM Assembly, Bash, SQL
Embedded: STM32 (bare-metal, register-level), ESP32, UART, SPI, I2C, DMA, interrupts, timers/PWM, logic analyzers, schematics
Robotics / CV: ROS2 (Humble/Iron), MAVLink, Gazebo, RPLIDAR, Jetson Orin Nano, YOLOv8/v11, PyTorch, TensorRT, ONNX, OpenCV, Roboflow
Tools: Linux, Git, GitHub Actions, Docker, Kubernetes, FastAPI, PostgreSQL, MongoDB, pytest, gtest